

The Δ -deformation, reduced

Consecutive Christoffel sampling, oscillation of ε , and why R_2 is not a Neumann remainder

Stefan Hamann

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Abstract

Rounds 381 and 385 left C_ε as the live first-order pairing $|\text{FO}_k[-\nu]| \leq \frac{1}{32}$ on the last twelve, written as $\gamma_k|Q_k - Q_{k-1}|$ with $Q_k = Q_k^T + \Delta_k$. This note asks whether a Szegő/Geronimus bound on the consecutive deformation $\Delta_k - \Delta_{k-1}$, together with second-order domination along $\mu \rightarrow \mu - \nu$, proves that box. What is proved: the split $\text{FO}_k = \gamma_k((Q_k^T - Q_{k-1}^T) + (\Delta_k - \Delta_{k-1}))$; homogeneity of monic Jacobi coefficients under a global mass scale; and that a Fejér-pure two-period mesh has $\varepsilon = 0$ and $\text{FO} = 0$. What is *not* proved: C_ε . The candidate $|\Delta_k - \Delta_{k-1}| \leq C \cdot \text{osc}(\varepsilon \text{ on } \nu)$ holds on MAIN with $C_* = 0.0056$ and is useless (the majorant $g(dQ^T + C_* \text{osc})$ reaches $0.191 > \frac{1}{32}$); consecutive Q is uncorrelated with $\|\varepsilon\|_\infty$ ($\rho = 0.025$) and with range-oscillation ($\rho = -0.287$). A scramble of the same d_{arm} channel keeps $\text{osc}(\varepsilon) = 3.54 < 8.76$ and sends $|\text{d}\Delta|$ to 10^6 . Along the μ -reference the midpoint ratio is 0.861 at degree 40 and 0.640 at full depth, both $> \frac{1}{4}$; $|R_2|/|\text{FO}| = 4.19$ on the last twelve. Neumann domination is false, so R_2 is not absorbed. The hydra-legal remainder is the r381 triple $G_\varepsilon^\mu \wedge C_\varepsilon \wedge R_2$, with C_ε a cancelled last-twelve pairing (not an ε -size bound) and G_ε^μ the Jacobi box of Fejér $\times d_{\text{arm}}$ (the r342 dictionary, $v_{\text{pred}} < 10^{-4}$, is the right language for that sibling). No RH claim.

Status. Lemmas 1–3 are proved (finite identities). Lemma 2 is exact over $\mathbb{Q}$. Proposition 1 is a finite machine census, not a cofinal law. Proposition 2 records that R_2 is not dominated on the μ -reference. Proposition 3 names the reduction. No RH claim. No G_ε claim. No C_ε claim.

1 The object

Write π_k for the μ -orthonormal polynomials and

$$Q_k := \langle \pi_k^2 \rangle_\nu, \quad Q_k^T := \frac{\sum_\nu v(1 + \cos 2k\theta)}{\sum_\mu w(1 + \cos 2k\theta)}, \quad \Delta_k := Q_k - Q_k^T. \quad (1)$$

Round 385: $\text{FO}_k[-\nu] = \gamma_k(Q_k - Q_{k-1})$ as an identity, and Q^T stays at the mass ratio α under the trivial Weyl bound. Condition C_ε of [2] is $|\text{FO}_k[-\nu]| \leq \frac{1}{32}$ for the last twelve degrees of the v_2 -chain. The mesh $x = \cos(2\pi j/L)$, $L = 4h - 2$, and the factorisation $w \propto (1 - x) d_{\text{arm}}$ are construction identities of [1]. Write $\varepsilon_j = (w_j/\bar{w}_j) - 1$ for the Fejér-stripped relative arm-length, $\bar{w} \propto (1 - x)$, mean-zero by construction.

The candidate of this round: a Geronimus bound $|\Delta_k - \Delta_{k-1}| \leq C \cdot \text{osc}(\varepsilon \text{ on } \nu)$ with an explicit C small enough that, together with the Chebyshev consecutive dQ^T and a dominated R_2 , one has C_ε .

2 What is exact

Lemma 1 (split). *On a quasi-definite atomic μ ,*

$$\text{FO}_k[-\nu] = \gamma_k \left((Q_k^T - Q_{k-1}^T) + (\Delta_k - \Delta_{k-1}) \right).$$

This is [3]'s pairing plus (1). FRAME-A $w = 9$, last twelve, floating-point residual $2 \cdot 10^{-15}$.

Lemma 2 (homogeneity). *If every weight of a quasi-definite atomic measure is scaled by a positive $\lambda \in \mathbb{Q}$, the monic orthogonal polynomials are unchanged and $\gamma_k = h_k/h_{k-1}$ is invariant. Exact over $\mathbb{Q}$ on a five-atom toy at $\lambda = 3$, depth 3.*

A global mass scale is therefore invisible to γ . The crude wall $\|\varepsilon\|_\infty = 5.166$ of [2] already quotients out that scale (Fejér-stripped, mean zero). What remains is variation of d_{arm} .

Lemma 3 (Fejér-pure). *On the two-period Fejér mesh of [4] (equal occupancy, weights $\propto (1-x)$, no d_{arm}), one has $\varepsilon = 0$, Q_k independent of k , and $\text{FO}_k = 0$. Checked at $S = 21$ and $S = 81$, amplitude $2/3$.*

This is the control deformation: when ε vanishes, consecutive Δ vanishes. The live field is d_{arm} .

3 Oscillation does not close C_ε

Write $\text{osc}_\nu(\varepsilon) = \max \varepsilon|_\nu - \min \varepsilon|_\nu$ and $C_{\text{emp}} = |\text{d}\Delta|_{12}/\text{osc}_\nu(\varepsilon)$.

Proposition 1 (census). *On the sealed constructors:*

1. *FRAME-A $w = 9$: last-twelve $|\text{FO}| = 0.00978 < \frac{1}{32}$, $\text{d}Q = 0.0387$, $\text{d}Q^T = 0.0197$, $\text{d}\Delta = 0.0495$, $\|\varepsilon_\mu\|_\infty = 5.166$, $\text{osc}_\nu = 8.764$, $C_{\text{emp}} = 0.0056$, $\gamma_{12} = 0.276$.*
2. *Core-42: $\text{corr}(\text{d}Q, \|\varepsilon\|_\infty) = 0.025$, $\text{corr}(\text{d}\Delta, \text{osc}_\nu) = -0.287$. Amplitude grows from 5 to 123 while consecutive Q stays $O(10^{-2})$ to $O(10^{-1})$. The L^∞ wall of [2] does not enter the consecutive difference — and neither does range-oscillation of ε (both quantities track the same d_{arm} dynamic range).*
3. *The osc-majorant $\gamma(\text{d}Q^T + C_* \text{osc}_\nu)$ with $C_* = 0.0056$ reaches $0.191 > \frac{1}{32}$ (closes on $5/42$). Doubling C_* already overshoots $\frac{1}{32}$ on $w = 9$ itself (majorant 0.0328).*
4. *Eleven small- N windows ($N \leq 220$): last-twelve $|\text{FO}| \leq 0.01182$ on MAIN and ≤ 0.01541 on χ_3 , both $< \frac{1}{32}$. (The 181-pack builder was not rerun; MAIN-85 at depth 80 is a fixed-depth smoothness census, not last-twelve of v_2 for large N , disclosed.)*

The bound $|\text{d}\Delta| \leq C_* \text{osc}_\nu$ is true on the source and does not prove C_ε : range-oscillation of ε is still $O(\|\varepsilon\|_\infty)$, and the last-twelve pairing cancels that amplitude. A single Fourier mode $a \cos(2m\theta)$ on an equal-weight cosine toy does move last-twelve FO proportionally to a ; the source d_{arm} is not that mode. The Geronimus picture that last-twelve $\delta\gamma$ tracks Fourier mode $q \sim M$ of d_{arm} remains the natural attack on C_ε and is not carried here — it is C_ε itself.

4 The μ -reference does not dominate R_2

Round 381 measured $|\text{quad}|/|\text{lin}| = 1.013$ along Fejér-constant $\rightarrow$ actual at degree 40. The reduction $G_\varepsilon \leftarrow G_\varepsilon^\mu \wedge C_\varepsilon \wedge R_2$ uses the other path: μ as reference, $-\nu$ as the finite step. A ratio $< \frac{1}{4}$ would have made R_2 a Neumann remainder of the Christoffel–Darboux perturbation of size $\text{mass}(\nu)/\text{mass}(\mu) = 0.061$.

Proposition 2 (R_2 not dominated). *On FRAME-A $w = 9$, writing $g(t)$ for last-twelve γ of $\mu - t\nu$:*

1. Degree 40: $|\text{quad}|/|\text{lin}| = 0.861$ (Fejér-path 1.013 reproduced).
2. Full depth $n = 182$: ratio 0.640; exact $|R_{2,k}|/|\text{FO}_k| = 4.19$ on the last twelve, with $R_{2,k} = \gamma_k(\mu - \nu) - \gamma_k^\mu - \text{FO}_k[-\nu]$.
3. Eleven small- N windows at degree 40: ratios in $[0.559, 1.126]$, all $> \frac{1}{4}$.

The mass ratio 0.06 is not the small parameter of the last twelve: the Christoffel sampling Q_k is $O(1)$ (last-twelve maximum 0.389). First-order perturbation theory about μ is a finite $O(1)$ step. R_2 is not absorbed.

5 Adversaries

Scramble. FRAME-A $w = 9$, seed 1: $\text{osc}_\nu(\varepsilon) = 3.54 < 8.76$ (smaller than MAIN) and $|\text{d}\Delta|_{12} = 2.16 \cdot 10^6$. The osc-bound with any C that works on MAIN is false on the scramble. Scramble does not inflate the d_{arm} field; it breaks the μ -OP sampling of ν (the r381/r385 named kill, reproduced as a kill of Geronimus-by-oscillation).

Mutant $C \times 2$. Twice the measured C_* already fails to prove C_ε on $w = 9$ (majorant $0.0328 > \frac{1}{32}$). Halving C_* fails on 6/42 core windows. The measured constant is not a closing constant.

Wrong reference. Replacing osc_ν by $\|\varepsilon\|_\infty$ makes the majorant worse (core maximum 0.353). Amplitude is the wrong norm for a consecutive difference.

Two-period, $c = 2/3$. Lemma 3: $\varepsilon = 0$ and $\text{FO} = 0$. C_ε holds trivially and $\lambda_{\max}(E_{22}) > 1$ still (the r385 witness that $Q < 1$ is not the floor plate). A control, not a kill of C_ε .

6 The reduction

Proposition 3 (reduction). *The candidate “oscillation of ε plus Neumann R_2 proves C_ε ” is strictly larger than the r381 triple:*

1. Split and homogeneity. SATZ (Lemmas 1–3). Consecutive Δ is a k -th layer pairing; a global mass scale does not enter.
2. Osc-Geronimus as a closing bound. False. The inequality with $C_* = 0.0056$ is a MAIN census, not a theorem of $\text{osc}(\varepsilon)$, and the implied majorant does not yield $\frac{1}{32}$. Scramble is the named kill.
3. Neumann R_2 . False on the μ -reference (Proposition 2). The reduction does not collapse to $G_\varepsilon^\mu \wedge C_\varepsilon$.
4. What remains. C_ε : cancelled last-twelve pairing of d_{arm} against $\pi_k^2 - \pi_{k-1}^2$ of μ (Geronimus/Fourier mode $q \sim M$, not $\text{osc}(\varepsilon)$). R_2 : Taylor remainder of the finite step $\mu \rightarrow \mu - \nu$. G_ε^μ : the Jacobi box of the positive measure $w \propto (1-x)d_{\text{arm}}$. The r342 dictionary $F_A(\xi) = -\log \pi + \Re \psi(\frac{1}{4} + i\xi/2)$ plus the two-cell tent, with measured $v_{\text{pred}} < 10^{-4}$, is the right language for that sibling — an explicit Jacobi perturbation of Jacobi-(0,1), not an oscillation bound.

The chain of [1, 2] stands: $V_3' \Leftarrow G_\varepsilon \Leftarrow G_\varepsilon^\mu \wedge C_\varepsilon \wedge R_2$.

T1 / COMPOSE unchanged. No RH claim.

7 Machine checks

Numbered lemmas: `verify_delta_deformation.py` (same directory). Standalone: homogeneity over $\mathbb{Q}$, one-Rayleigh, Fejér-pure two-period, Neumann bar, Chebyshev–Gauss, split toy. Construction: FRAME-A $w = 9$ pins, split residual, R_2 midpoint ratios, $C \times 2$ mutant, scramble kill, two-period control. Discovery census: `delta_deformation_probe.py` (`--smoke` / `full`). Exit line: DELTA DEFORMATION VERIFIED.

Claim boundary

Finite identities for Jacobi perturbation and Christoffel sampling on a fixed node set, a named reduction of the osc-Geronimus candidate for C_ε , a scramble break of that candidate, and a midpoint record that R_2 is not dominated on the μ -reference. No statement about zeros of $\zeta(s)$, in either direction. No RH claim.

References

- [1] S. Hamann, *The V'_3 -lemma, reduced*, standalone note, August 2026, `rh/problem/v3prime_proof.tex`.
- [2] S. Hamann, *The G_ε -lemma, reduced*, standalone note, August 2026, `rh/problem/g-eps_lemma.tex`.
- [3] S. Hamann, *Christoffel quietness, reduced*, standalone note, August 2026, `rh/problem/christoffel_quiet.tex`.
- [4] S. Hamann, *The entry lemma of the pivot band*, standalone note, August 2026, `rh/problem/pivot_entry_lemma.tex`.